

# Report 1 - Gradient Descent

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## 1 Algorithm

This algorithm uses gradient descent to minimize the function  $y=x^2$ . It starts from an initial value of  $x$  and updates the value step by step. In each iteration,  $x$  is adjusted using the gradient  $2x$ . The learning rate controls how big each step is. Over time, the value of  $x$  gets closer to 0, where the function reaches its minimum. Here the result with initial value 10 and learning rate 0.01

Table 1: Result table by time

| Time | $x$  | $f(x)$ |
|------|------|--------|
| 1    | 8.00 | 64.00  |
| 2    | 6.40 | 40.96  |
| 3    | 5.12 | 26.21  |
| 4    | 4.10 | 16.78  |
| 5    | 3.28 | 10.74  |
| 6    | 2.62 | 6.87   |
| 7    | 2.10 | 4.40   |
| 8    | 1.68 | 2.81   |
| 9    | 1.34 | 1.80   |
| 10   | 1.07 | 1.15   |

## 2 Effect of learning rates

The table below shows the results of running the test with different learning rates.

Table 2: Result table by time

| Time | $lr = 0.01$ |       | $lr = 0.1$ |       | $lr = 0.2$ |       | $lr = 0.4$ |      |
|------|-------------|-------|------------|-------|------------|-------|------------|------|
|      | x           | f(x)  | x          | f(x)  | x          | f(x)  | x          | f(x) |
| 1    | 9.80        | 96.04 | 8.00       | 64.00 | 6.00       | 36.00 | 2.00       | 4.00 |
| 2    | 9.60        | 92.24 | 6.40       | 40.96 | 3.60       | 12.96 | 0.40       | 0.16 |
| 3    | 9.41        | 88.58 | 5.12       | 26.21 | 2.16       | 4.67  | 0.08       | 0.01 |
| 4    | 9.22        | 85.08 | 4.10       | 16.78 | 1.30       | 1.68  | 0.02       | 0.00 |
| 5    | 9.04        | 81.71 | 3.28       | 10.74 | 0.78       | 0.60  | 0.00       | 0.00 |
| 6    | 8.86        | 78.47 | 2.62       | 6.87  | 0.47       | 0.22  | 0.00       | 0.00 |
| 7    | 8.68        | 75.36 | 2.10       | 4.40  | 0.28       | 0.08  | 0.00       | 0.00 |
| 8    | 8.51        | 72.38 | 1.68       | 2.81  | 0.17       | 0.03  | 0.00       | 0.00 |
| 9    | 8.34        | 69.51 | 1.34       | 1.80  | 0.10       | 0.01  | 0.00       | 0.00 |
| 10   | 8.17        | 66.76 | 1.07       | 1.15  | 0.06       | 0.00  | 0.00       | 0.00 |

Here are the comments.

- A small learning rate leads to slow but stable convergence, often requiring many iterations.
- A large learning rate can speed up training but may cause overshooting or divergence.
- Choosing an appropriate learning rate is crucial for balancing convergence speed and stability.